Multiscale Global Context Network for Semantic Segmentation of High-Resolution Remote Sensing Images

Qiaolin Zeng, Jingxiang Zhou, Jinhua Tao, Liangfu Chen, Xuerui Niu, and Yumeng Zhang

***Abstract*— Semantic segmentation of high-resolution remote sensing images (HRSIs) is a challenging task because objects in HRSIs usually have great scale variance and appearance variance. Although deep convolutional neural networks (DCNNs) have been widely applied in the semantic segmentation of HRSIs, they have inherent limitations in capturing global context. Attention mechanisms and transformer can effectively model long-range dependencies, but they often result in high computational costs when being applied to process HRSIs. In this article, an encoder–decoder network (MSGCNet) is proposed to fully and efficiently model multiscale context and long-range dependencies of HRSIs. Specifically, the multiscale interaction (MSI) module employs an efficient cross-attention to facilitate interaction among multiscale features of the encoder, which bridges the semantic gap between high- and low-level features and introduces more scale information to the network. In order to efficiently model long-range dependencies in both spatial and channel dimensions, the transformer-based decoder block (TBDB) implements window-based efficient multihead self-attention (W-EMSA) and enables interactions cross windows. Furthermore, to further integrate the global context generated by TBDB, the scale-aware fusion (SAF) module is proposed to deeply supervise the decoder, which iteratively fuses hierarchical features through spatial attention. As demonstrated by both quantitative and qualitative experimental results on two publicly available datasets, the proposed MSGCNet exhibits superior performance compared to currently popular methods. The code will be available at http://github.com/JingxiangZhou/MSGCNet.**

***Index Terms*— Attention mechanism, remote sensing images, semantic segmentation, transformer.**

I. Introduction

IN THE past few years, the advancements in remote sensing technologies have provided a large number of high-resolution remote sensing images (HRSIs) with abundant spatial details and semantic information. Semantic segmentation of HRSIs aims to assign a specific category to each pixel, which plays a crucial role in enhancing our understanding of geographical environments. Therefore, semantic segmentation of HRSIs has been applied to many fields, such as land use detection [1], [2], urban planning [3], [4], [5], building extraction [6], [7], [8], and road extraction [9], [10]. However, objects in HRSIs have the characteristics of great scale variance, high intraclass variability, and low interclass variability [11]. Although traditional methods [12], [13], [14] based on feature engineering achieve pixel-by-pixel classification by manually designing features and classifiers, it is difficult to cope with the above challenges due to their limited features extraction capabilities.

Recently, deep convolutional neural networks (DCNNs) have overcome the shortcomings of traditional methods due to their ability of automatically extracting hierarchical and abstract feature representations. FCN [15] is the first DCNN to achieve end-to-end semantic segmentation, but its simple upsampling operation leads to the loss of spatial information. Therefore, U-Net [16] based on the encoder–decoder framework uses skip connections in the decoding stage to fuse the encoder features at the corresponding scale to retain more spatial information. To augment the multiscale representation capability of the network, DeepLabV3+ [17] uses atrous convolutions to construct an atrous spatial pyramid pooling (ASPP) module, which aggregates context from receptive fields at multiple scales. Although the theoretical receptive fields of these networks are already larger than the input images, the empirical receptive fields of DCNNs have been proven to be much smaller than the theoretical receptive fields [18]. In addition, the size of the receptive field can roughly indicate the degree of utilization of contextual information in DCNNs [19]. As a result, the aforementioned methods solely rely on convolution to integrate semantic features from different levels, which restricts their capability to adaptively interact multiscale semantic and model global context.

An effective technology for modeling global information is attention mechanism, which can be divided into self-attention and scaling attention. A typical example of self-attention is DANet [20], which designs the position and channel attention modules to adaptively model spatial and channel dimensional global context, respectively. Its position attention module models the correlation between each pixel and

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all other pixels, and each pixel aggregates features based on its correlation with all pixels through matrix operations. Nevertheless, the computational complexity of the network will increase sharply as the input image size grows due to the $O(N^2)$ complexity of standard self-attention. The scaling attention aims to learn attention weights to highlight or suppress features of specific dimensions. For example, the squeeze-and-excitation (SE) module of SENet [21] models the dependency between channel descriptors and assigns weights to each channel for calibrating features. Compared with the SE module, the convolutional block attention module (CBAM) [22] allocates the attention weight in channel and spatial dimensions successively through a tandem structure, while the bottleneck attention module (BAM) [23] is designed as a parallel structure. However, if these attention mechanisms are combined simply, it will result in redundant attention weights due to the complex features of HRSIs, which limits the performance of networks.

Vision transformer (ViT) [24] is the first transformer-based model in image processing tasks, which opens a new prospect with its strong long-range dependencies modeling ability. Compared with ViT, Pyramid ViT (PVT) [25] overcomes the shortcoming that ViT can only produce fixed resolution and nonfine-grained features, so it is also suitable for dense prediction tasks. By limiting multihead self-attention (MSA) in local windows and designing the window shift mechanism, Swin Transformer [26] can effectively model long-range dependencies while reducing the computational cost. Because of the absolute position encoding, ViT can only process sequences with fixed length. Therefore, ResT [27] and conditional positional encoding for ViT (CPVT) [28] embed the relative and absolute position information of the patch into tokens through convolution operations, which allows the network to adaptively encode the position of input images with different sizes. Moreover, ResT designs a convolution-based stem module at the beginning of the network to compensate for the lack of local information in subsequent transformer layers. Although transformer has greater potential, many transformer-based networks perform patch embedding on input images, which makes the extracted low-level features insufficient, especially for HRSIs. In addition, these transformer-based models require more memory and computational cost due to MSA, which limits their application potential in processing of HRSIs.

Therefore, to address the problem of underutilizing multiscale information and capturing long-range dependencies with high computational complexity, a multiscale global context network (MSGCNet) is proposed for semantic segmentation of HRSIs. Specifically, the ResNet34 [29] is used as the encoder to effectively extract local context. Considering that the traditional skip connection ignores the semantic gap between the low-level features of the encoder and the high-level features of the decoder, and it only implicitly fuses features of different resolutions, we developed the multiscale interaction (MSI) module, which bridges the semantic gap between multilevel features through the attention mechanism and explicitly embeds more multiscale information into features. In addition, we modify efficient attention [30] to cross-attention to improve the interaction efficiency of MSI module. For the decoder, we developed a novel transformer block for modeling long-range dependencies of features, which is called transformer-based decoder block (TBDB). Compared with traditional window-based MSA (W-MSA), the window-based efficient multihead self-attention (W-EMSA) of TBDB further reduces the complexity and also embeds channel information into attention. Also, a concise cross window interaction (CWI) module is proposed for interacting information between windows. Moreover, the multilayer perceptron with convolution (C-MLP) module is designed to enhance the local feature capturing ability of TBDB. Finally, we design the scale-aware fusion (SAF) module to further exploit the multiscale global context output by TBDB and deeply supervise the decoder, which iteratively integrates decoded features through a spatial attention mechanism. In summary, the main contributions of this article are as follows.

1) An MSI module integrated efficient cross-attention is designed to explicitly facilitate the fusion process of multiscale semantics to replace traditional skip connections.
2) We propose a novel TBDB for modeling long-range dependencies of features. The W-EMSA module in TBDB has lower complexity by applying efficient MSA and channel attention within local windows. In addition, with the proposed CWI module and C-MLP module, TBDB can achieve full interaction across windows and learning of local features.
3) We propose the SAF module, which uses fine-grained spatial attention to iteratively integrate the global context of the decoder to enhance the discriminability of the final prediction.
4) A novel deep learning framework (MSGCNet) is proposed, which achieves excellent performance in semantic segmentation of HRSI on two public datasets thanks to its modeled multiscale long-range dependencies.

II. Related Works

A. Multiscale Context in Semantic Segmentation of HRSIs

The great scale variance of objects in HRSIs requires networks to have the ability of capturing multiscale context. Many methods propose the corresponding modules to explicitly extract multiscale features, and most of these modules are composed of convolutional layers with different receptive fields in parallel. For example, MACU-Net [31] designs an asymmetric convolutional block to capture fine features and adaptively fuses features through multiscale skip connections. HFGNet [32] develops a parallel convolution module with multiple kernel sizes to extract multiscale presentation and implements feature interaction through dense connections. Similarly, MAFANet [33] uses parallel strip convolutions instead of standard convolutions at the end of encoder to extract multiscale features. However, for large objects that require larger receptive fields, adopting such modules is inefficient because the parameters and computational costs grow exponentially with the convolution kernel size [34]. Although U-Net implicitly fuses features of different resolutions through skip connections, it lacks further effective representation of

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multiscale information. Therefore, some other networks extract multiscale information by enhancing the fusion of features at different stages. Different from the structure of U-Net, CFNet [35] concatenates the features of two adjacent layers of the encoder and fuses them through convolution to generate multiscale representation. SSAtNet [36] takes into account the hierarchical dependency between multiscale features in the parallel structure of ASPP. Thus, it implements progressive aggregation from large receptive field to small receptive field features through convolution. MPSegNet [37] designs three subnetworks that focus on segmenting objects at three scales, and the multiscale features output by these subnetworks are further integrated by learning weights in the spatial dimension. However, the multiscale features extracted by these methods are not interactive enough, which makes the network insufficiently utilize multiscale information.

B. Attention Mechanism in Semantic Segmentation of HRSIs

For the class of a single pixel, the methods based solely on convolutions only consider its correlation with surrounding local pixels. Therefore, many methods introduce attention mechanisms in DCNNs to model global information or highlight beneficial features. MAResU-Net [38] and MANet [39] introduce a comparable parallel self-attention module, which serves as a substitute for the skip connection. LANet [40] uses patch-wise local attention to aggregate contextual information, which embeds the local attention into features to bridge the semantic gap during the fusion process. SAPNet [41] designs a synergistic attention module, which alleviates the bias caused by using attention separately in the two dimensions. To capture fine-grained features that are more sensitive to the category of objects, HMANet [42] develops a class-augmented attention module to model the dependencies between specific categories and channels. CaFE [43] leverages efficient channel attention to enhance the feature fusion capability between two modalities belonging to the same class, which helps the network capture more significant feature representations. Similarly, AFNet [44] combines spatial and channel dimensional scaling attention to fuse multimodal features. ABCNet [45] utilizes spatial attention to fuse features of the spatial path and contextual path, and this integration enables the final feature representation to encompass abundant spatial details and contextual information. In order to refine the features to be sent by skip connections, many attention-based methods only apply attention mechanisms on single-scale features [35], [38], [39]. Since the interactions between features at different resolutions are ignored, they do not improve the utilization of multiscale semantics, which limits the performance of networks. In addition, dot-product attention brings excessive computational cost to the network [20], [46], which makes it difficult for them to achieve a better tradeoff between efficiency and performance.

C. Transformer in Semantic Segmentation of HRSIs

With the great success of ViT in image processing, more and more methods have applied transformer to semantic segmentation of HRSIs [47], [48]. Since transformers are more focused on modeling long-range dependencies and DCNNs are better at capturing low-level features of images, most methods are based on hybrid architectures of transformer and DCNNs. For example, BANet [49] combines features generated by a dependency path utilizing ResT to effectively capture long-range dependencies, along with a texture path leveraging stacked convolutional blocks to extract fine-grained features. ST-UNet [50] embeds global information of Swin Transformer into another convolution-based path, which forms a dual encoding architecture. To suppress background noise and enhance foreground saliency, RSSFormer [51] designs a transformer-based adaptive fusion module that fuses multiscale features at each stage of HRNet [52]. EMRT [47] employs a spatial pooling branch to extract multiscale information, which serves as the query of MSA to model multiscale global contextual feature representation. However, the decoders of these methods are far less strong than the encoders, and the stacked transformer blocks in the encoder bring a heavy computational burden. UNetFormer [53] develops efficient global–local attention to couple the global and local information in the feature decoding process, which replaces the MSA with high complexity in standard transformer blocks. Similarly, CMTFNet [54] develops a multiscale transformer block for decoding. It takes the output of the multiscale branch based on atrous convolution as the key and value of MSA, which enhances the multiscale representation capability in long-range dependency modeling. However, these methods use additionally developed modules to process high-resolution features at the end of the decoder in order to refine the features of the final output [53], [54]. They do not apply the developed transformer block to higher resolution features in the decoder. Although this approach can reduce the complexity of the network, it leads to insufficient modeling of the long-range dependencies of the features used for prediction.

III. Methodology

In this section, the overall structure of MSGCNet is presented and it mainly includes the MSI module, the TBDB, and the SAF module.

A. Network Architecture

In order to take advantage of both the characteristics of DCNNs that are better at capturing local features and the characteristics of transformer that is better at modeling long-range dependencies, our network adopts a hybrid architecture of DCNNs and transformer. As shown in Fig. 1, MSGCNet uses ResNet34 [29] as the encoder to fully extract the low-level features of the image, which contains four-stage Resblocks. Also, each stage of the encoder downsamples the features at a ratio of 2. The multiscale features output by the first three stages of the encoder are input to the MSI module, and the output is connected to the decoder. The TBDB is used at different stages of the decoder to obtain global contextual features with different resolutions. Finally, the SAF module adaptively integrates multiscale features from each stage of the decoder, which cooperates with the segmentation head to output the final prediction.

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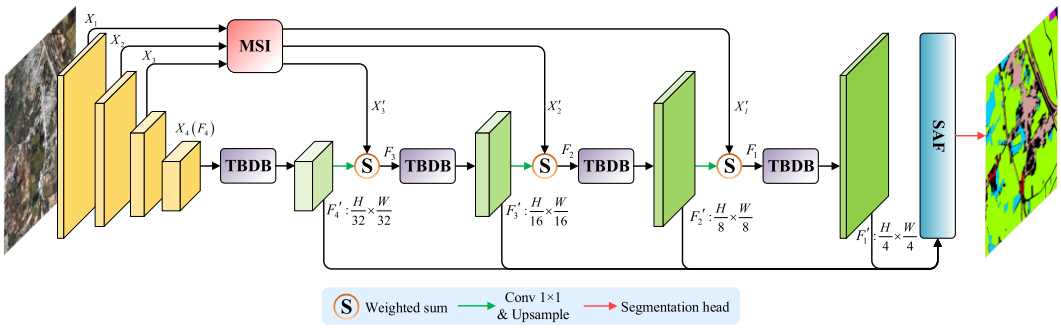


Fig. 1. Overall framework of the proposed MSGCNet.

B. MSI Module

In the encoder–decoder structure, the downsampling operation of the encoder results in the inevitable loss of spatial details in low-resolution features [55]. The decoder gradually upsamples the low-resolution features to the size of the original image, and the lost spatial information is compensated by skip connections. However, the encoder features are lower level because they are computed in earlier layers of the network. In contrast, the features of the decoder have higher levels because they have undergone more processing in the deeper layers of the network [34]. Therefore, ignoring the semantic gap between high- and low-level features makes traditional skip connections introduce irrelevant noise [56], which is not beneficial to the fusion of multiscale features in the encoder–decoder network. To solve this problem, the MSI module is proposed to bridge the semantic gap and explicitly embed more scale information into the features.

The higher resolution features of the encoder contain more spatial information, so we feed the features from the first three stages of the encoder into the MSI module to generate multiscale features with spatial details, and their sizes are $H/4 \times W/4$, $H/8 \times W/8$, and $H/16 \times W/16$. As shown in Fig. 2(a), the MSI module adaptively interacts with the multiscale hierarchical features through the cross-attention mechanism. Compared with ordinary self-attention that can only be implemented on features of a single scale, cross-attention can improve the utilization of features at different resolutions. In addition, since dot-product attention has a complexity of $O(N^2)$, we develop efficient cross-attention based on efficient attention [30] to improve the efficiency of interaction, which has a linear complexity. For a single-scale feature, we utilize efficient cross-attention to query its correlations with other-scale features and aggregate multiscale context, thereby introducing more scale information to the network.

Specifically, the process of MSI module is summarized as follows:

$$X_1', X_2', X_3' = \mathrm{MSI}(X_1, X_2, X_3) \tag{1}$$

where X_1, X_2, and X_3 are the inputs of the MSI module, and X_1', X_2', and X_3' are the outputs. For $X_i \in \mathbb{R}^{N_i \times C_i}$, we use 1×1 convolution layers to obtain query matrixes Q_i^j and Q_i^k, the key matrix K^i, and the value matrix V^i

$$Q_i^j, Q_i^k, K^i, V^i = \mathrm{Conv1} \times 1(X_i) \tag{2}$$

where $i, j, k \in \{1, 2, 3\}, i \neq j \neq k$, $Q_i^j \in \mathbb{R}^{N_i \times D_{\text{key}}^j}$, $Q_i^k \in \mathbb{R}^{N_i \times D_{\text{key}}^k}$, $K^i \in \mathbb{R}^{N_i \times D_{\text{key}}^i}$, and $V^i \in \mathbb{R}^{N_i \times D_{\text{value}}^i}$. N_i and C_i are the number of pixels and channels of X_i, respectively. For $V^i \in \mathbb{R}^{N_i \times D_{\text{value}}^i}$, $D_{\text{value}}^i = C_i$. Assuming the common setting of $D_{\text{key}}^i = C_i$, we can further use a parameter μ to adjust the number of dimensions D_{key}^i of $K^i \in \mathbb{R}^{N_i \times D_{\text{key}}^i}$, $Q_j^i \in \mathbb{R}^{N_j \times D_{\text{key}}^i}$, and $Q_k^i \in \mathbb{R}^{N_k \times D_{\text{key}}^i}$

$$D_{\text{key}}^i = \frac{C_i}{\mu}. \tag{3}$$

The reduction of D_{key}^i can reduce the computational cost of the matrix operation of efficient cross-attention, which further improves its efficiency compared to dot-product cross-attention. The experiment in Section V-A2 shows that when μ is 4, the MSI module can achieve the best tradeoff between efficiency and performance. For $i, j \in \{1, 2, 3\}, i \neq j$, the efficient cross-attention can be defined as follows:

$$X_i^j = \mathrm{ECA}\left(Q_i^j, K_j, V_j\right) = \rho_{\text{row}}\left(Q_i^j\right)\left(\rho_{\text{col}}\left(K_j\right)^{\mathrm{T}} V_j\right) \tag{4}$$

where ECA denotes efficient cross-attention, which converts the matrix multiplication order in dot-product attention from $(QK^{\mathrm{T}})V$ to $Q(K^{\mathrm{T}}V)$, and this conversion has almost no impact on the effect and reduces the complexity from $O(N^2)$ to $O(D_{\text{key}} D_{\text{value}})$ [30]. ρ_{row} and ρ_{col} represent applying the softmax function to each row and column of the matrix, respectively. Finally, X_i, X_i^j, and X_i^k have the same spatial size, and they are concatenated and fed into a 1×1 convolution layer to generate X_i'

$$X_i' = \mathrm{Conv1} \times 1\left(\mathrm{Concat}\left(X_i, X_i^j, X_i^k\right)\right) \tag{5}$$

where $i, j, k \in \{1, 2, 3\}$, and they are pairwise unequal. At each stage of the decoder, we align the features F_{i+1}' output by TBDB to the shape of the features X_i' output by the MSI module through 1×1 convolution and upsampling operations. We then use the weight sum operation to fuse these two features with the same shape. Specifically, weighted sum is defined as follows:

$$F_i = \alpha \cdot X_i' + (1 - \alpha) \cdot \mathrm{Conv1} \times 1\&\mathrm{Up}\left(F_{i+1}'\right) \tag{6}$$

where Conv1 $\times$ 1&Up represents the 1×1 convolution and upsampling operation and α is a learnable parameter. F_i represents the fused feature, which is fed into the next TBDB, and $i \in \{1, 2, 3\}$. The weighted sum operation adaptively

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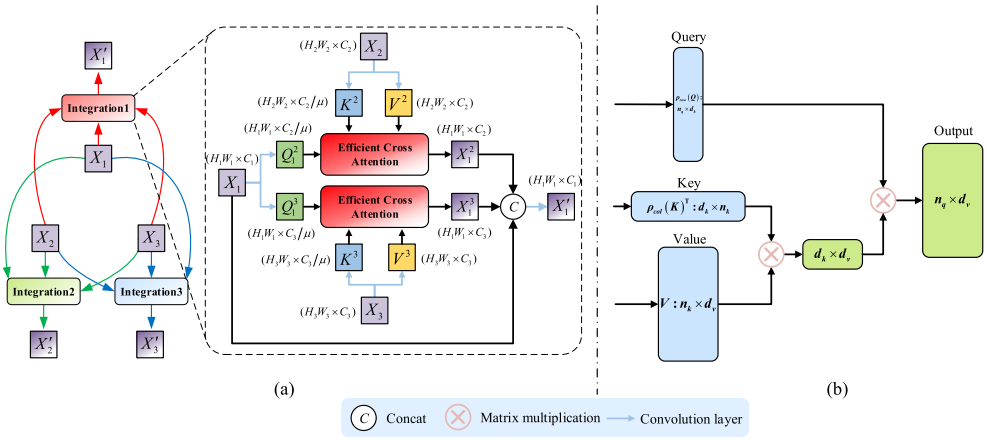


Fig. 2. (a) Structure of the MSI module. (b) Details of efficient attention.

assigns weights to features according to their contribution to segmentation accuracy, thereby learning more generalizable fusion features [53].

C. TBDB

The MSA for modeling the dependencies of tokens and the multilayer perceptron (MLP) for extracting features of input are two key components of the standard transformer block. The computational complexity of MSA in ViT [24] is $O(N^2)$, which means that when the size of the input image increases, the computational cost of the network will increase dramatically. Therefore, Swin Transformer [26] limits the calculation of MSA to local windows, which greatly improves the efficiency of the network. However, the computational complexity of W-MSA of Swin Transformer still scales squarely with the size of windows, which means that larger window sizes still cause huge computational burden [57]. To further improve the efficiency of W-MSA, based on efficient attention, we developed the W-EMSA module, which implements multihead efficient self-attention within local windows. In addition, channel attention is also embedded in the W-EMSA module, which enables TBDB to capture global information from both spatial and channel dimensions simultaneously.

As shown in Fig. 3, TBDB consists of a position encoding module, a W-EMSA module, a CWI module, a C-MLP module, two BatchNorm (BN) layers, and two residual connections.

First, the TBDB utilizes pixel-attention (PA) in ResT [27] to encode position, which has variable length according to the input tokens. PA obtains weights of pixels through a 3 × 3 depth-wise convolution layer and embeds the position encoding into the input features through element-wise multiplication after applying the sigmoid function for scaling

$$X = \mathrm{PA}(X_{\mathrm{in}}) = X_{\mathrm{in}} \odot \sigma(\mathrm{DWConv3} \times 3(X_{\mathrm{in}})) \tag{7}$$

where X_{in} denotes the input feature, $\odot$ represents element-wise multiplication, σ represents the sigmoid function, and DWConv3 × 3 represents the 3 × 3 depth-wise convolution.

After encoding position, the feature is partitioned into nonoverlapping windows. For the ith attention head of the

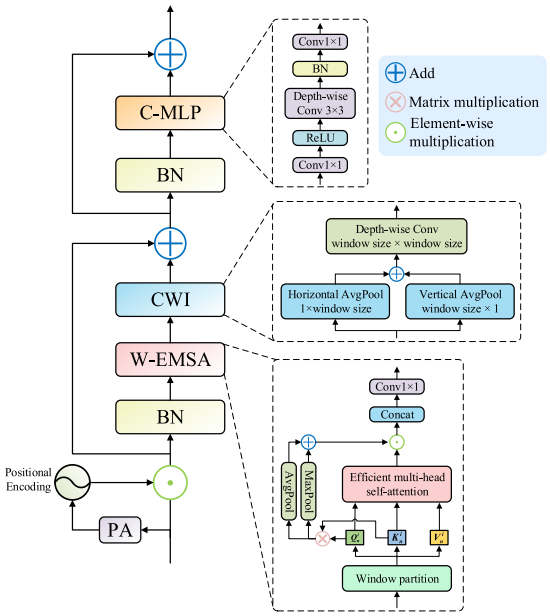


Fig. 3. Structure of the TBDB.

nth window, the self-attention input (Q_n^i, K_n^i, V_n^i) is obtained through three linear layers

$$Q_n^i = X_n W_Q^i, K_n^i = X_n W_K^i, V_n^i = X_n W_V^i \tag{8}$$

where $W_Q^i, W_K^i \in \mathbb{R}^{C\times(C/4I)}$, $W_V^i \in \mathbb{R}^{C\times(C/I)}$, $Q_n^i, K_n^i \in \mathbb{R}^{S^2\times(C/4I)}$, $V_n^i \in \mathbb{R}^{S^2\times(C/I)}$, $X_n \in \mathbb{R}^{S^2\times C}$, C denotes the number of channels, S denotes the window size, and I represents the number of self-attention heads. The formula for performing efficient MSA within windows can be summarized as follows:

$$\mathrm{head}_n^i = \mathrm{EA}\left(Q_n^i, K_n^i, V_n^i\right) \odot \mathrm{CA}\left(Q_n^i, K_n^i\right) \tag{9}$$

$$\mathrm{CA}\left(Q_n^i, K_n^i\right) = \mathrm{maxpool}\left(\rho_{\mathrm{row}}\left(Q_n^i\right)^{\mathrm{T}} \rho_{\mathrm{col}}\left(K_n^i\right)\right) \oplus \mathrm{avgpool}\left(\rho_{\mathrm{row}}\left(Q_n^i\right)^{\mathrm{T}} \rho_{\mathrm{col}}\left(K_n^i\right)\right) \tag{10}$$

$$\mathrm{W-EMSA}(X_n) = \mathrm{Concat}\left(\mathrm{head}_n^1, \ldots, \mathrm{head}_n^I\right) W^O \tag{11}$$

where EA denotes the efficient attention for a certain attention head, which is calculated as (4). CA denotes the channel

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self-attention. Specifically, the product of transpose of the normalized Q_n^i and normalized K_n^i is summed through the maximum pooling layer and the average pooling layer output, respectively. $\oplus$ represents the element-wise summation and $W^O \in \mathbb{R}^{C\times C}$.

Swin Transformer is designed with an additional shifted transformer block to model cross-window correlations, which results in more computational costs. Therefore, we designed a more concise CWI module to fully interact the context between windows with strip pooling and large kernel convolution. As shown in Fig. 3, the features output by W-EMSA are input into average pooling layers with kernel sizes of $1 \times S$ and $S \times 1$, which interact between windows with a stride of 1 in the horizontal and vertical directions, respectively. The features output by two pooling layers are summed and further fused through an $S \times S$ depth-wise convolution.

Finally, considering that transformer is better at modeling long-range dependencies, we add a 3×3 depth-wise convolutional layer into the standard MLP, which can enhance the local information capturing capability of TBDB. In addition, BN is applied before the W-EMSA module and the C-MLP module, and the residual connection is applied after the CWI module and the C-MLP module.

D. SAF Module

The motivation of deep supervision is to integrate features from different levels to obtain the best final prediction, thereby better perceiving scale variance of objects [56]. Many networks have been deeply supervised with the help of feature fusion [53], [56], [58], which can provide timely feedback for the feature learning of networks to enhance the discriminability of the final prediction. However, the features fused directly at different levels to deeply supervise the network may weaken the multiscale representation ability because of the semantic gap between these features [59]. Therefore, inspired by CA-Net [60], the SAF module adaptively learns weights for pixels through spatial attention to finely integrate the global context of the decoder in an iterative manner.

Specifically, as shown in Fig. 4, the number of channels of features output by TBDB is unified to 64 to reduce the computational complexity. For two adjacent features, the low-resolution feature is upsampled to the size of the high-resolution feature, and they are fused through the scale-aware (SA) block [59]. The two features F_A and F_B input into the SA block are concatenated and fed into the convolution and softmax layer to obtain two spatial attention maps $A', B' \in \mathbb{R}^{H\times W}$. The SA block uses the learned A' and B' to adaptively fuse F_A and F_B

$$F_{\text{fusion}} = A' \odot F_A + B' \odot F_B. \tag{12}$$

The fused features are upsampled and then iteratively fused with higher resolution features through the SA block. Finally, the features generated by the SAF module are fed into the segmentation head to generate the prediction. The segmentation head is composed of a 3×3 convolution layer, a 1×1 convolution layer, and an upsampling operation.

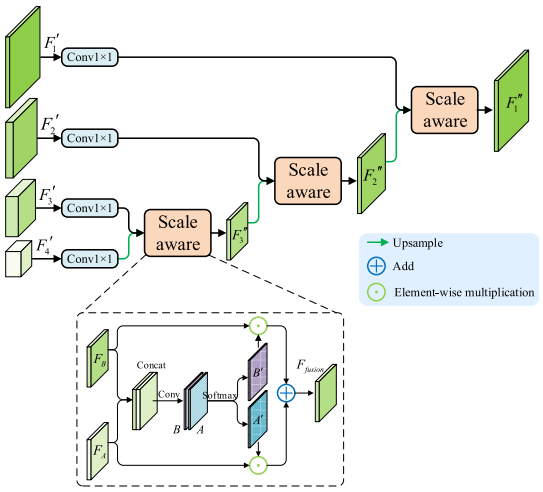


Fig. 4. Structure of the SAF module.

IV. Experimental Settings

A. Datasets

1) Beijing Land-Use (BLU) Dataset: The BLU [61] dataset comprises four large HRSIs in the suburbs of Beijing, which were collected by the Beijing-2 satellite in June 2018. The images contain three bands of red, green, and blue (RGB) and six semantic categories with a ground sampling distance (GSD) of 0.8 m. It has 196 images for training, 28 for validation, and 32 for testing. Also, the size of each image is 2048×2048.

2) Gaofen Image Dataset (GID): The GID [62] dataset contains 150 RGB images with a GSD of 0.8 m and 16 categories, which were captured by the Gaofen-2 satellite. Since the training and testing set divisions of this dataset are not provided, this study cropped its images to a pixel size of 2048×2048 and randomly divided it into 112 images for training, 16 for validation, and 32 for testing.

When inputting images into the model for training, the images are cropped to a size of 512×512 with a step of 256. In addition, random flipping and random rotation are used for data augmentation during training.

B. Implementation Details

All experiments are implemented on an RTX 3090 GPU with the PyTorch framework. The batch size is set to 8, and the Adam optimizer is used for training. The initial learning rate is set to 0.0001, which is updated according to the poly strategy as follows:

$$l_r = 0.0001 \times \left(1 - \frac{\text{iterations}}{\text{total_interations}}\right)^{\text{power}} \tag{13}$$

where the power is set to 1.5. The maximum number of epochs is 100, and the early stopping strategy is used with a patience of 10. Moreover, the total loss used in this study is as follows:

$$L_{\text{Total}} = L_{\text{CE}} + 0.5 \times L_{\text{Dice}} + 0.5 \times L_{\text{Focal}} \tag{14}$$

where L_{CE} denotes the cross-entropy loss and L_{Focal} and L_{Dice} denote the focal loss and dice loss, respectively, which can

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TABLE I
ABLATION EXPERIMENTAL RESULTS OF OVERALL FRAMEWORK ON THE BLU AND GID DATASETS

Dataset	MSI	TBDB	SAF	OA (%)	mF1 (%)	mIoU (%)
BLU				82.90	75.70	62.19
	✓			85.94	80.15	67.92
	✓	✓		86.76	81.81	70.13
	✓	✓	✓	87.16	82.34	70.85
GID				79.73	62.01	49.35
	✓			80.35	63.41	51.36
	✓	✓		81.88	64.80	52.79
	✓	✓	✓	82.08	65.85	53.58

effectively alleviate the negative impact of category imbalance on training.

This study uses overall accuracy (OA), mean F1 score (mF1), and mean intersection over union (mIoU) to evaluate the performance of networks, and they are defined as follows:

$$\text{OA} = \frac{\text{TP}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{15}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{16}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{17}$$

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{18}$$

$$\text{mIoU} = \frac{\text{TP}}{\text{TP} + \text{FN} + \text{FP}} \tag{19}$$

where TP, TN, FP, and FN represent the true positive, true negative, false positive, and false negative, respectively. In addition, floating-point operations (FLOPs) are used to evaluate the complexity. The memory footprint and the number of parameters are used to evaluate the memory requirements of the model.

V. EXPERIMENTAL RESULTS

A. Ablation Study

In this section, ablation experiments of the overall framework are implemented on the BLU and GID datasets. The remaining more ablation experiments are performed on the BLU dataset.

1) Ablation of the Overall Architecture: The ablation experimental results of the overall framework proposed by us are shown in Table I. On the BLU dataset, compared with the baseline network built by ResNet34 combined with upsampling operations, the MSI module introduces more scale information into the network, which increases OA, mF1, and mIoU by 3.04%, 4.45%, and 5.73%, respectively. The TBDB further improves OA, mF1, and mIoU by 0.82%, 1.66%, and 2.21%, respectively, which is attributed to the long-range dependencies that TBDB models. The SAF module improves OA, mF1, and mIoU to 87.16%, 82.34%, and 70.85%, respectively, by adaptively aggregating the multiscale information in the decoder. On the GID dataset, the MSI module, the TBDB, and the SAF module increase the mIoU of the baseline network to 51.36%, 52.79%, and 53.58%, respectively, which verifies the effectiveness of these modules.

2) Reduction Ratio of D_{key} in the MSI Module: In the MSI module, we use efficient cross-attention for the interaction between multiscale features. For μ in (2), when it increases, the computational cost of efficient cross-attention can be further reduced. Therefore, the MSI module is added in the baseline network to conduct experiments on the value of μ. As shown in Table II, when μ is 1, the MSI module can achieve 86.15% of OA, 80.26% of mF1, and 68.09% of mIoU. When μ is 2, the mF1 only decreases slightly by 0.03%, but the FLOPs, memory usage, and the number of parameters of the MSI module are reduced by nearly 26%, 16%, and 18%, respectively, compared with μ of 1. When μ is 4, the FLOPs, memory usage, and the number of parameters are reduced by 39%, 24%, and 26%, respectively, compared to μ of 1, but it has comparable performance as μ of 1. When μ is 8, the information exchanged by the MSI module is significantly reduced, so the mIoU decreases by 1.72% compared with μ of 1, and the further decrease of the FLOPs, memory usage, and the number of parameters is not significant. Therefore, this study sets μ in (2) to 4, which can make the MSI module achieve the optimal balance between efficiency and performance.

3) Superiority of the MSI Module: In the MSI module, for a single feature, it queries its correlations with features of other resolutions through cross-attention, thereby integrating more scale information. Since the features of three layers of the encoder are input into the MSI module, the efficient cross-attention is implemented a total of six times. To further illustrate the superiority of the MSI module, we replace efficient cross-attention in the MSI module with dot-product cross-attention. As shown in Table III, using ordinary dot-product cross-attention to interact with multiscale features achieves 85.93%, 80.02%, and 67.75% performance on OA, mF1, and mIoU, respectively. However, the FLOPs and memory requirement of the MSI module are as high as 559918.45 M and 779.69 MB. When using efficient cross-attention, the mIoU is improved by 0.17% compared to dot-product cross-attention, which achieves comparable performance. It is remarkable that the FLOPs and memory usage with efficient cross-attention are only about 13% and 14% of dot-product cross-attention, respectively, which significantly improves the efficiency of MSI.

4) Superiority of the W-EMSA Module: We take the last TBDB in the decoder as an example, the size of its input is 128 × 128. Table IV illustrates the superiority of W-EMSA. As the window size increases, FLOPs and memory usage of W-EMSA do not increase further. When the feature map cannot be completely partitioned by a window of a specific size, the size of the feature map will be padded to an integer multiple of the window size. For example, when the window size is 8, a feature map of size 128 × 128 can be completely partitioned into 256 windows. When the window size is 6, the feature map will be padded to a size of 132 × 132 to be partitioned into 484 windows. Therefore, when the size of the feature map is not divisible by the size of the window, computing W-EMSA on the padded feature map results in slightly larger memory requirements and FLOPs. In contrast, the FLOPs required by W-MSA increase sharply as

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TABLE II
ABLATION EXPERIMENTAL RESULTS OF D_{KEY} IN THE MSI MODULE ON THE BLU DATASET

μ	OA (%)	mF1 (%)	mIoU (%)	FLOPs (M)	Memory (MB)	Parameter (M)
1	86.15	80.26	68.09	11895.05	140.16	0.84
2	86.01	80.23	67.97	8820.62	118.08	0.69
4	85.94	80.15	67.92	7283.41	106.73	0.62
8	85.88	79.09	66.37	6514.80	101.30	0.58

TABLE III
COMPARISON OF DOT-PRODUCT CROSS-ATTENTION AND THE EFFICIENT CROSS-ATTENTION ON THE BLU DATASET

Attention mechanism	OA (%)	mF1 (%)	mIoU (%)	FLOPs (M)	Memory (MB)	Parameter (M)
Dot-product cross-attention	85.93	80.02	67.75	55918.46	779.69	0.62
Efficient cross-attention	85.94	80.15	67.92	7283.41	106.73	0.62

the window size increases, and it also requires more memory than W-EMSA. When the window size is 16 × 16, W-EMSA can achieve the optimal tradeoff between computational cost and performance, and its FLOPs and memory are only 37% and 49% of W-EMSA, respectively. Therefore, this study sets the window size of W-EMSA module of TBDB to 16 × 16.

5) Ablation of the TBDB: The TBDB is mainly composed of the W-EMSA module, CWI module, and CMLP module. To verify their effectiveness, the TBDB is further added to the baseline network combined with the MSI module. As can be seen in Table V, adding the W-EMSA module improves the mF1 and mIoU by 0.61% and 0.74%, respectively. The CWI module interacts with information between windows to better capture global information, which further increases mIoU by 0.82%. The C-MLP module is an important component of TBDB, which aims to extract feature representation. Also, it increases OA, mF1, and mIoU to 86.76%, 81.81%, and 70.13%, respectively.

6) Superiority of the SAF Module: In the decoder, the SAF module is proposed to fuse multiscale global contextual features and deeply supervise the network. We compare the SAF module with the SAF module without SA blocks. As shown in Table VI, for the baseline network combined with MSI module and TBDB, the SAF module without SA blocks only improves the mIoU of the network to 70.41%, which demonstrates that simply aggregating the multiscale features of the decoder has a limited effect because there are semantic gaps between these features. The SAF module adaptively assigns weights to pixels of multiscale features through spatial attention, which improves OA, mF1, and mIoU to 87.16%, 82.34%, and 70.85%, respectively.

B. Comparison of State-of-the-Art Methods

To verify the superiority of MSGCNet, we compare it with some popular methods on the BLU and GID datasets. The selected methods mainly include convolution-based networks (e.g., U-Net [16], SegNet [63], DeepLabV3+ [17], and PSPNet [19]), attention mechanism-based networks (e.g., DANet [20], CCNet [64], MAResU-Net [38], MACU-Net [31], MANet [39], CFNet [35], and A^2-FPN [46]), and transformer-based methods (e.g., Swin-Unet [65], UNetFormer [53], BuildFormer [57], and BANet [49]).

1) Comparison With Other Methods on the BLU Dataset: Table VII reports the quantitative comparison results on the BLU dataset. The OA, mF1, and mIoU of the proposed MSGCNet reach the highest 87.16%, 82.34%, and 70.85%, respectively. Compared with PSPNet, which does not adaptively capture multiscale context, MSGCNet improves mIoU by 1.93%. Compared with DANet, MSGCNet achieves a 3.11% improvement in mIoU, which indicates that attention-based networks for segmentation of natural image may struggle to cope with the challenges of HRSIs. Moreover, the OA, mF1, and mIoU of MSGCNet are improved by 0.53%, 0.68%, and 1.01%, respectively, compared with MAResU-Net. In addition, MSGCNet achieves the highest IoU in four semantic categories, which benefits from fully captured multiscale global contextual features.

Fig. 5 shows the qualitative results of MSGCNet and some methods on the BLU dataset. In the first example, for the buildings within the dotted box, the segmentation results of MSGCNet have clearer boundaries than other methods. In the second example, U-Net incorrectly labels the water category within the dotted box as agricultural or vegetation, whereas the segmentation result produced by MSGCNet is more closely to the ground truth. In the third example, the visual features of the water category are very close to the agricultural and vegetation categories, and MSGCNet also accurately identifies it.

2) Comparison With Other Methods on the GID Dataset: Table VIII reports the quantitative comparison results on the GID dataset. The OA, mF1, and mIoU of the proposed MSGCNet reach the highest 82.08%, 65.85, and 53.58%, respectively. Compared with CCNet and CFNet, the mIoU of MSGCNet is improved by 0.79% and 0.95%, respectively. A^2-FPN only sends global information through traditional skip connections, and MSGCNet improves its mIoU by 2.94%. Compared with OA, mF1, and mIoU of BANet, MSGCNet improved them by 0.42%, 1.67%, and 1.46%, respectively. Moreover, MSGCNet reaches the highest IoU for three semantic categories, and the IoU of the remaining categories also achieves comparable performance to other methods.

Fig. 6 shows the visualization results on the GID dataset. In the first example, except for MSGCNet, the remaining methods segment the irrigated land category in the top-right corner into the background category to varying degrees. In the second example, the rural residential category in the center

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TABLE IV
ABLATION EXPERIMENTAL RESULTS OF THE WINDOW SIZE OF W-EMSA MODULE ON THE BLU DATASET

Global context	Window size	OA (%)	mF1 (%)	mIoU (%)	FLOPs (M)	Memory (MB)	Parameter (K)
W-EMSA	6	86.53	81.09	69.21	388.59	32.46	10.40
	8	86.40	80.63	68.53	373.36	26.76	
	10	86.74	81.69	69.95	380.87	31.11	
	12	86.92	81.40	69.53	388.50	31.27	
	14	86.70	80.97	68.94	420.25	33.91	
	16	86.76	81.81	70.13	373.31	26.28	
W-MSA	6	86.03	80.44	68.24	448.69	32.73	10.40
	8	86.21	80.50	68.35	503.31	30.13	
	10	86.76	81.47	69.67	612.28	40.43	
	12	86.29	80.76	68.66	749.77	47.95	
	14	86.63	80.52	68.35	989.72	59.66	
	16	86.37	81.06	69.14	1006.63	54.13	

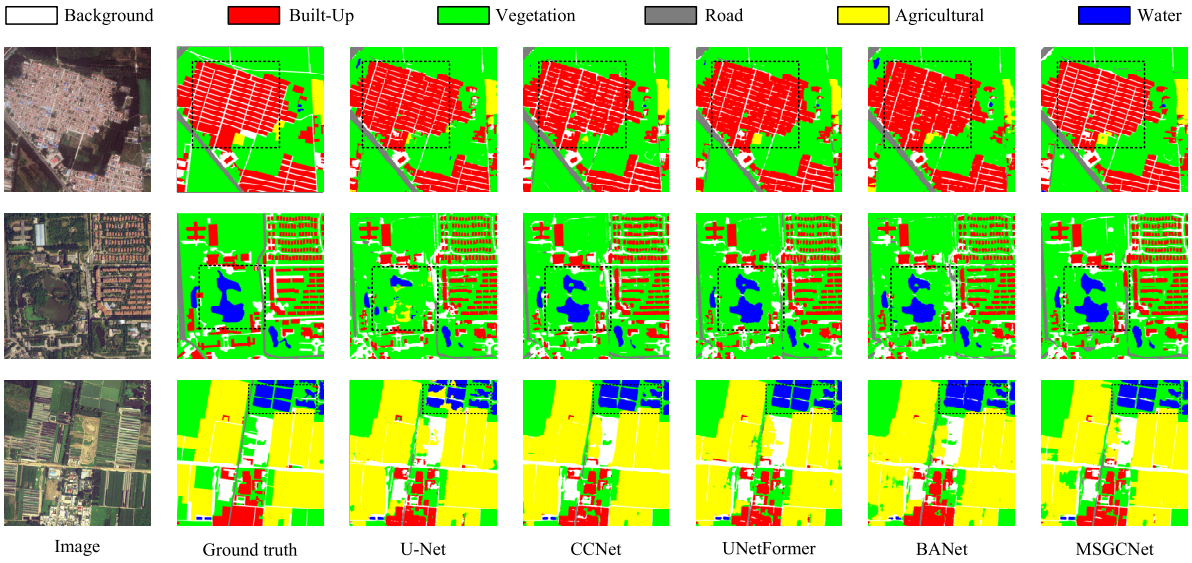


Fig. 5. Qualitative comparison with state-of-the-art methods on the BLU dataset.

TABLE V
ABLATION EXPERIMENTAL RESULTS OF THE TBDB ON THE BLU DATASET

W-EMSA	CWI	C-MLP	OA (%)	mF1 (%)	mIoU (%)
			85.94	80.15	67.92
✓			86.14	80.76	68.66
✓	✓		86.48	81.35	69.48
✓	✓	✓	86.76	81.81	70.13

TABLE VI
ABLATION EXPERIMENTAL RESULTS OF THE SAF MODULE ON THE BLU DATASET

Fusion method	OA (%)	mF1(%)	mIoU (%)
SAF w/o SA	87.02	82.03	70.41
SAF	87.16	82.34	70.85

of the image is accurately segmented by MSGCNet, closely resembling the ground truth. Conversely, the other methods exhibit significant misjudgments in this area. In the last example, MSGCNet also achieves the best segmentation result.

C. Feature Visualization

As shown in Fig. 7, we use the Grad-CAM-based [66] method to visualize the features iteratively fused in the SAF module, which are the multiscale global context output by TBDB. In the first example, we use the water category as the specified category in class activation maps, and the segmentation mask of the water category is also visualized. The first SA block fuses the global information output by the first and second TBDB, and there is still noise in the visualized features. As more global information output by the remaining TBDB is integrated in the second and third SA blocks, the network strengthens the correlations between pixels of the water category, and the noise of features becomes less. In the second example, as the features output by TBDB are continuously fused in the SAF module, the noise in the feature map and the number of misjudged pixels continue to decrease.

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TABLE VII
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE BLU DATASET

Method	Backbone	Per-Category IoU						OA	mF1	mIoU
		Background	Built-Up	Vegetation	Road	Agricultural	Water			
U-Net	-	55.28	77.56	82.50	49.13	75.69	70.21	86.24	80.59	68.39
SegNet	VGG16	54.20	75.23	81.68	51.42	74.51	69.16	85.56	80.19	67.70
DeepLabV3+	ResNet34	49.42	76.08	81.23	50.07	72.21	72.28	84.78	79.45	66.88
PSPNet	ResNet34	56.58	77.10	82.71	51.02	77.23	68.86	86.55	81.03	68.92
DANet	ResNet34	54.43	75.52	81.07	49.53	75.81	70.09	85.48	80.17	67.74
CCNet	ResNet34	52.74	76.51	79.43	48.71	70.63	72.35	84.29	79.42	66.73
MAResU-Net	-	55.84	77.71	82.64	**51.84**	76.92	74.11	86.63	81.66	69.84
MACU-Net	-	54.90	77.39	82.36	49.29	74.67	67.87	86.09	80.14	67.75
MANet	ResNet34	54.62	76.05	81.66	49.33	76.37	72.05	85.90	80.56	68.35
CFNet	ResNet34	52.88	76.63	82.14	48.54	76.38	74.53	85.93	80.59	68.52
A^2-FPN	ResNet34	54.57	77.00	82.16	51.17	75.05	74.03	86.01	81.06	68.99
Swin-Unet	Swin-T	55.06	75.69	81.73	48.33	75.98	68.31	85.69	79.97	67.52
UNetFormer	ResNet34	55.53	76.76	82.15	47.63	75.76	73.53	86.04	80.66	68.56
BuildFormer	-	52.96	76.89	82.25	51.14	75.35	68.19	85.91	80.19	67.80
BANet	ResT-Lite	55.75	77.26	**82.95**	48.85	78.04	73.16	86.70	81.21	69.33
MSGCNet (ours)	ResNet34	**58.16**	**79.15**	82.94	50.94	**78.29**	**75.64**	**87.16**	**82.34**	**70.85**

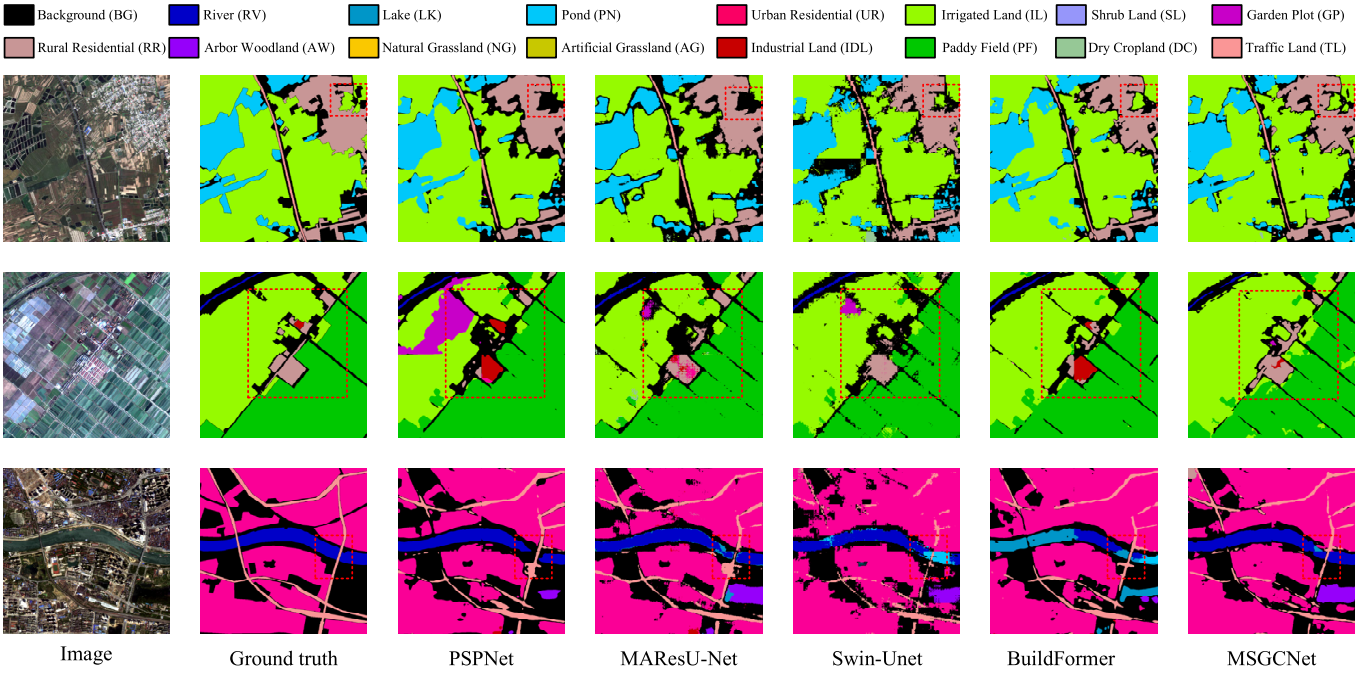


Fig. 6. Qualitative comparison with state-of-the-art methods on the GID dataset.

In the third example, as the SA block continues to integrate the multiscale global information, the boundaries of the areas identified by the agricultural category become clearer and clearer, and the correlations between pixels become closer, too.

D. Complexity Comparison of Different Models

Table IX shows the computational complexity, the number of parameters, and the inference speed of all methods under the same operating environment. As can be seen from Table IX, the model with the lowest computational complexity is UNetFormer because it only applies the transformer block to three lower resolution features in the decoder, and the number of channels of its decoded features is uniformly reduced. However, these operations limit the decoding ability of UNetFormer, and its mIoU is 2.29% lower than that of MSGCNet. The method with the lowest number of parameters is MACU-Net, but its simple structure and less parameters limit its feature representation capabilities. Thus, the mIoU of

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TABLE VIII
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS ON THE GID DATASET

Method	Backbone	Per-Category IoU																OA	mF1	mIoU
		BG	IDL	UR	RR	TL	PF	IL	DC	GP	AW	SL	NG	AG	RV	LK	PN			
U-Net	-	70.80	53.18	65.61	46.34	49.46	71.49	69.25	25.91	11.74	66.73	0	38.89	11.51	91.89	42.50	77.91	80.55	61.76	49.58
SegNet	VGG16	67.61	49.72	60.58	42.40	45.41	72.27	66.26	21.84	9.54	64.26	0	34.80	13.43	84.19	41.19	77.55	77.95	59.51	46.94
DeepLabV3+	ResNet34	70.11	55.65	65.79	43.36	48.48	80.45	72.18	30.23	16.10	66.28	4.76	37.00	12.89	**93.67**	58.85	78.49	80.79	64.42	52.14
PSPNet	ResNet34	71.69	56.40	63.28	49.16	51.69	73.50	70.01	**41.27**	7.64	63.27	0	33.95	**16.20**	91.07	55.58	77.76	80.66	63.58	51.40
DANet	ResNet34	70.56	56.51	67.29	46.06	49.57	76.65	71.63	33.47	14.42	66.57	6.58	35.92	13.56	90.67	60.78	77.37	81.09	64.83	52.35
CCNet	ResNet34	70.81	55.32	66.55	48.26	49.79	76.60	70.74	32.81	16.09	65.82	0	39.83	15.61	90.12	**68.75**	77.57	81.09	64.96	52.79
MAResU-Net	-	71.29	55.74	66.55	47.62	50.49	77.15	68.75	24.33	21.29	64.02	0	**40.62**	12.61	92.42	53.67	75.96	80.96	63.73	51.41
MACU-Net	-	69.21	50.85	65.12	44.51	46.52	78.28	66.76	27.65	10.88	62.80	0	37.29	12.20	86.61	41.31	74.39	79.19	60.87	48.40
MANet	ResNet34	70.11	55.18	66.92	40.39	39.58	70.10	70.15	29.52	0	62.54	0	37.09	0	90.57	49.08	72.15	80.10	58.39	47.09
CFNet	ResNet34	71.41	57.37	65.66	48.74	47.11	79.01	71.11	31.97	13.80	64.86	5.60	36.39	16.16	91.19	62.47	**79.29**	81.29	65.03	52.63
A^2-FPN	ResNet34	71.13	51.47	65.28	44.95	49.82	76.88	71.10	38.62	**25.22**	**67.23**	5.56	34.04	14.22	86.40	35.49	72.82	80.70	63,79	50.64
Swin-Unet	Swin-T	70.40	49.82	65.00	40.72	39.97	**80.88**	68.84	37.03	10.95	63.94	0	30.99	5.49	89.11	40.11	69.76	79.85	59.82	47.69
UNetFormer	ResNet34	70.57	54.23	66.59	47.78	49.45	75.27	70.05	26.82	10.73	66.03	0	32.83	13.38	89.14	62.99	75.06	80.78	62.73	50.68
BuildFormer	-	71.09	55.97	**69.04**	47.96	47.61	74.28	69.46	34.85	10.50	66.48	0	38.57	9.91	90.95	50.64	74.56	81.06	62.83	50.74
BANet	ResT-Lite	71.36	**58.44**	66.63	**50.39**	48.14	74.39	**73.45**	38.08	13.93	66.87	0.04	34.47	12.08	91.89	56.27	77.57	81.66	64.18	52.12
MSGCNet (ours)	ResNet34	**72.20**	58.40	68.31	46.59	**51.85**	80.26	71.51	29.00	20.80	65.33	**6.85**	39.67	11.16	91.69	64.99	78.62	**82.08**	**65.85**	**53.58**

TABLE IX
COMPARISON OF THE COMPLEXITIES OF OUR METHOD WITH OTHER METHODS

Method	FLOPs (G)	Parameter (M)	Speed (FPS)	mIoU
U-Net	321.46	17.26	62	68.39
SegNet	322.79	29.45	58	67.70
DeepLabV3+	63.62	26.12	**113**	66.88
PSPNet	319.66	27.57	57	68.92
DANet	198.64	22.78	91	67.74
CCNet	193.68	22.78	86	66.73
MAResU-Net	70.06	26.28	50	69.84
MACU-Net	67.21	**5.15**	32	67.75
MANet	63.77	22.10	67	68.35
CFNet	43.97	22.93	109	68.52
A^2-FPN	161.15	43.04	97	68.99
SwinU-Net	89.88	34.63	40	67.52
UNetFormer	**21.43**	21.79	60	68.56
BuildFormer	234.30	38.35	32	67.80
BANet	30.39	12.73	57	69.33
MSGCNet (ours)	58.43	27.02	46	**70.85**

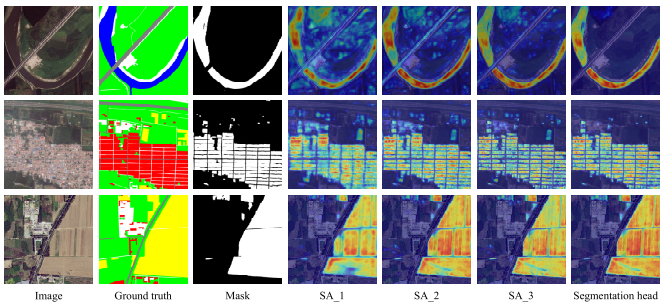


Fig. 7. Visualization of output features in the SAF module. Here, "SA_1," "SA_2," and "SA_3" represent the first, second, and third scale aware blocks in the SAF module, respectively.

MSGCNet is 3.1% higher than it. Due to the advantages of CNN, the speed of most CNN-based methods is faster than transformer-based methods. Among them, DeepLabV3+ has the fastest speed, but its mIoU is 3.97% lower than MSGCNet. In general, compared with other models, MSGCNet achieves the best balance between computational complexity, parameter size, and speed, which achieves the best performance.

VI. CONCLUSION

In this article, we propose the MSGCNet for semantic segmentation of HRSIs. MSGCNet employs efficient cross-attention to build the MSI module, which adaptively interacts with multiscale features. In the decoder, we implement efficient MSA within windows to develop the TBDB that can efficiently capture spatial and channel dimensional long-range dependencies. In addition, we propose a spatial attention-based SAF module, which integrates the multiscale global contextual features in the decoder. These modules are highly portable and can be easily plugged into other models. The extensive ablation studies and comparative experiments on two public datasets demonstrate the effectiveness of our MSGCNet, which provides a new perspective for the application of attention mechanism and transformer in processing of HRSIs.

REFERENCES

[1] R. Li, S. Zheng, C. Duan, L. Wang, and C. Zhang, "Land cover classification from remote sensing images based on multi-scale fully convolutional network," *Geo-Spatial Inf. Sci.*, vol. 25, no. 2, pp. 278–294, Jan. 2022.

[2] B. Chen, M. Xia, and J. Huang, "MFANet: A multi-level feature aggregation network for semantic segmentation of land cover," *Remote Sens.*, vol. 13, no. 4, p. 731, Feb. 2021.

[3] R. Cao et al., "Deep learning-based remote and social sensing data fusion for urban region function recognition," *ISPRS J. Photogramm. Remote Sens.*, vol. 163, pp. 82–97, May 2020.

[4] H. Luo, C. Chen, L. Fang, K. Khoshelham, and G. Shen, "MS-RRFSegNet: Multiscale regional relation feature segmentation network for semantic segmentation of urban scene point clouds," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8301–8315, Dec. 2020.

[5] H. Bi, L. Xu, X. Cao, Y. Xue, and Z. Xu, "Polarimetric SAR image semantic segmentation with 3D discrete wavelet transform and Markov random field," *IEEE Trans. Image Process.*, vol. 29, pp. 6601–6614, 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:59:20 UTC from IEEE Xplore. Restrictions apply.

[6] M. D. Hossain and D. Chen, "A hybrid image segmentation method for building extraction from high-resolution RGB images," *ISPRS J. Photogramm. Remote Sens.*, vol. 192, pp. 299–314, Oct. 2022.

[7] H. Jung, H.-S. Choi, and M. Kang, "Boundary enhancement semantic segmentation for building extraction from remote sensed image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5215512.

[8] W. Li et al., "Joint semantic-geometric learning for polygonal building segmentation from high-resolution remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 201, pp. 26–37, Jul. 2023.

[9] L. Dai, G. Zhang, and R. Zhang, "RADANet: Road augmented deformable attention network for road extraction from complex high-resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, doi: 10.1109/TGRS.2023.3237561.

[10] Q. Xu, C. Long, L. Yu, and C. Zhang, "Road extraction with satellite images and partial road maps," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, doi: 10.1109/TGRS.2023.3261332.

[11] F. I. Diakogiannis, F. Waldner, P. Caccetta, and C. Wu, "ResUNet-A: A deep learning framework for semantic segmentation of remotely sensed data," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 94–114, Apr. 2020.

[12] D. Li, S. Chen, C. Feng, W. Li, and K. Yu, "Bias correction of intensity inhomogeneous images hybridized with superpixel segmentation," *Biomed. Signal Process. Control*, vol. 71, Jan. 2022, Art. no. 103207.

[13] A. Radman, N. Zainal, and S. A. Suandi, "Automated segmentation of iris images acquired in an unconstrained environment using HOG-SVM and GrowCut," *Digit. Signal Process.*, vol. 64, pp. 60–70, May 2017.

[14] S. Qi, J. Ma, J. Lin, Y. Li, and J. Tian, "Unsupervised ship detection based on saliency and S-HOG descriptor from optical satellite images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 7, pp. 1451–1455, Jul. 2015.

[15] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[16] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput. Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.

[17] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with Atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[18] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Object detectors emerge in deep scene CNNs," 2014, *arXiv:1412.6856*.

[19] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.

[20] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3141–3149.

[21] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 8, pp. 2011–2023, Aug. 2020.

[22] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.

[23] J. Park, S. Woo, J.-Y. Lee, and I. So Kweon, "BAM: Bottleneck attention module," 2018, *arXiv:1807.06514*.

[24] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[25] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 548–558.

[26] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[27] Q. Zhang and Y. Yang, "ResT: An efficient transformer for visual recognition," 2021, *arXiv:2105.13677*.

[28] X. Chu, Z. Tian, B. Zhang, X. Wang, and C. Shen, "Conditional positional encodings for vision transformers," 2021, *arXiv:2102.10882*.

[29] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[30] S. Zhuoran, Z. Mingyuan, Z. Haiyu, Y. Shuai, and L. Hongsheng, "Efficient attention: Attention with linear complexities," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 3530–3538.

[31] R. Li, C. Duan, S. Zheng, C. Zhang, and P. M. Atkinson, "MACU-Net for semantic segmentation of fine-resolution remotely sensed images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[32] Z. Wang, S. Zhang, C. Zhang, and B. Wang, "Hidden feature-guided semantic segmentation network for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5603417.

[33] K. Chen, M. Xia, H. Lin, and M. Qian, "Multiscale attention feature aggregation network for cloud and cloud shadow segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, doi: 10.1109/TGRS.2023.3283435.

[34] N. Ibtehaz and M. S. Rahman, "MultiResUNet: Rethinking the U-Net architecture for multimodal biomedical image segmentation," *Neural Netw.*, vol. 121, pp. 74–87, Jan. 2020.

[35] C. Peng, K. Zhang, Y. Ma, and J. Ma, "Cross fusion net: A fast semantic segmentation network for small-scale semantic information capturing in aerial scenes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2021.3053062.

[36] Q. Zhao, J. Liu, Y. Li, and H. Zhang, "Semantic segmentation with attention mechanism for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2021.3085889.

[37] R. Hang, P. Yang, F. Zhou, and Q. Liu, "Multiscale progressive segmentation network for high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2022.3207551.

[38] R. Li, S. Zheng, C. Duan, J. Su, and C. Zhang, "Multistage attention ResU-Net for semantic segmentation of fine-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[39] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607713.

[40] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.

[41] X. Li et al., "A synergistical attention model for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5400916.

[42] R. Niu, X. Sun, Y. Tian, W. Diao, K. Chen, and K. Fu, "Hybrid multiple attention network for semantic segmentation in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2021.3065112.

[43] A. Zheng, J. He, M. Wang, C. Li, and B. Luo, "Category-wise fusion and enhancement learning for multimodal remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2022.3225843.

[44] X. Yang et al., "An attention-fused network for semantic segmentation of very-high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 238–262, Jul. 2021.

[45] R. Li, S. Zheng, C. Zhang, C. Duan, L. Wang, and P. M. Atkinson, "ABCNet: Attentive bilateral contextual network for efficient semantic segmentation of fine-resolution remotely sensed imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 181, pp. 84–98, Nov. 2021.

[46] R. Li, L. Wang, C. Zhang, C. Duan, and S. Zheng, "A^2-FPN for semantic segmentation of fine-resolution remotely sensed images," *Int. J. Remote Sens.*, vol. 43, no. 3, pp. 1131–1155, Feb. 2022.

[47] T. Xiao, Y. Liu, Y. Huang, M. Li, and G. Yang, "Enhancing multiscale representations with transformer for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605116.

[48] C. Zhang, W. Jiang, Y. Zhang, W. Wang, Q. Zhao, and C. Wang, "Transformer and CNN hybrid deep neural network for semantic segmentation of very-high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408820.

[49] L. Wang, R. Li, D. Wang, C. Duan, T. Wang, and X. Meng, "Transformer meets convolution: A bilateral awareness network for semantic segmentation of very fine resolution urban scene images," *Remote Sens.*, vol. 13, no. 16, p. 3065, Aug. 2021.

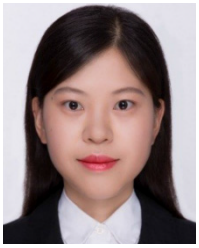
[50] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.

[51] R. Xu, C. Wang, J. Zhang, S. Xu, W. Meng, and X. Zhang, "RSSFormer: Foreground saliency enhancement for remote sensing land-cover segmentation," *IEEE Trans. Image Process.*, vol. 32, pp. 1052–1064, 2023.

[52] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 03:59:20 UTC from IEEE Xplore. Restrictions apply.

[53] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, 2022.
[54] H. Wu, P. Huang, M. Zhang, W. Tang, and X. Yu, "CMTFNet: CNN and multiscale transformer fusion network for remote-sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2004612.
[55] Q. Zeng, J. Zhou, and X. Niu, "Cross-scale feature propagation network for semantic segmentation of high-resolution remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.
[56] Y. Sun, D. Dai, Q. Zhang, Y. Wang, S. Xu, and C. Lian, "MSCA-Net: Multi-scale contextual attention network for skin lesion segmentation," *Pattern Recognit.*, vol. 139, Jul. 2023, Art. no. 109524.
[57] L. Wang, S. Fang, X. Meng, and R. Li, "Building extraction with vision transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5625711.
[58] Z. Zhou, M. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Proc. Int. Workshop Deep Learn. Med. Image Anal.*, vol. 11045, 2018, pp. 3–11.
[59] S. Feng et al., "CPFNet: Context pyramid fusion network for medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 39, no. 10, pp. 3008–3018, Oct. 2020.
[60] R. Gu et al., "CA-Net: Comprehensive attention convolutional neural networks for explainable medical image segmentation," *IEEE Trans. Med. Imag.*, vol. 40, no. 2, pp. 699–711, Feb. 2021.
[61] L. Ding et al., "Looking outside the window: Wide-context transformer for the semantic segmentation of high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, doi: 10.1109/TGRS.2022.3168697.
[62] X.-Y. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322.
[63] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.
[64] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.
[65] H. Cao et al., "Swin-Unet: Unet-like pure transformer for medical image segmentation," 2021, *arXiv:2105.05537*.
[66] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.



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